Bayesian Model Development for Hydroformylation Reaction Optimization

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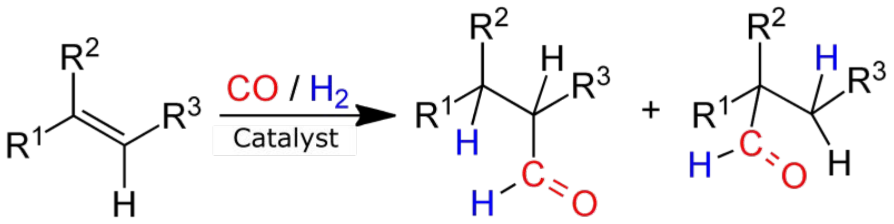
Introduction

Hydroformylation, is a process for the production of *aldehydes* from *alkenes.*



It is an important and powerful tool for the chemical industry since *Aldehydes* can be easily converted into a wide variety of high-value products, from detergents to drugs.

Oxo synthesis research focuses in increasing production capacity while overcoming common challenges like *selectivity, yield,* material and energy consumption.

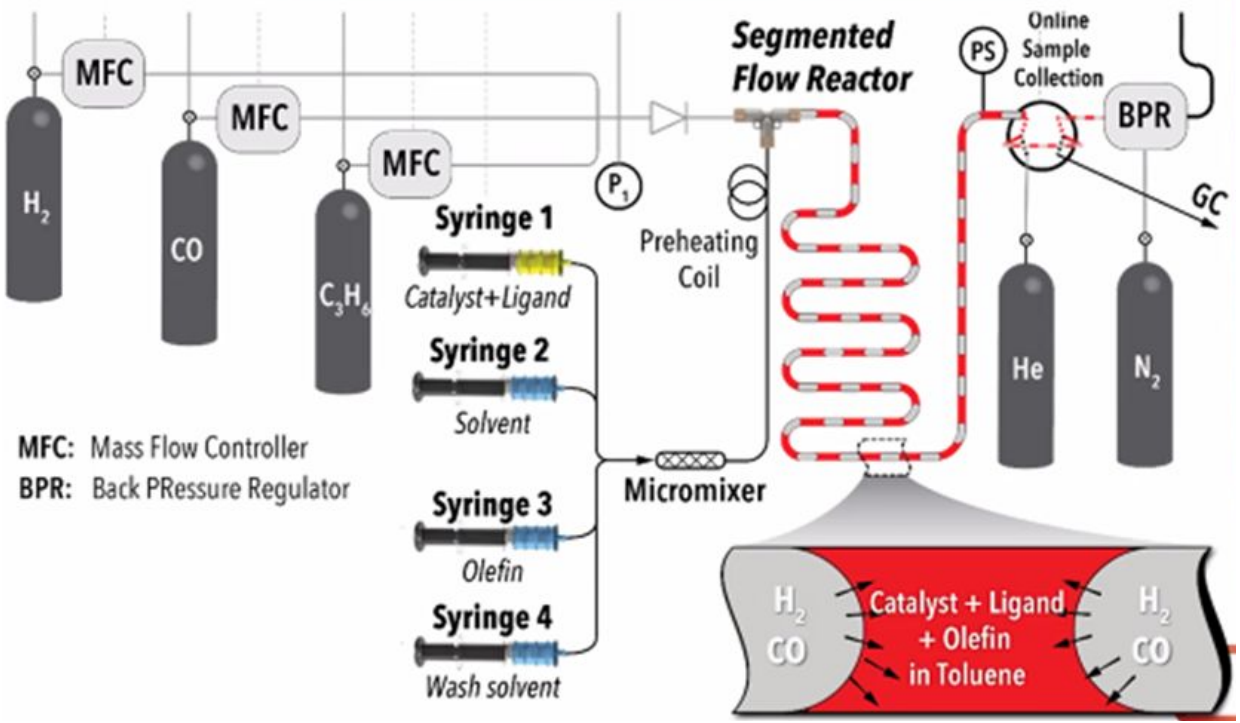
Description of Data

Microfluidics facilitate the exploration of the parameter space:

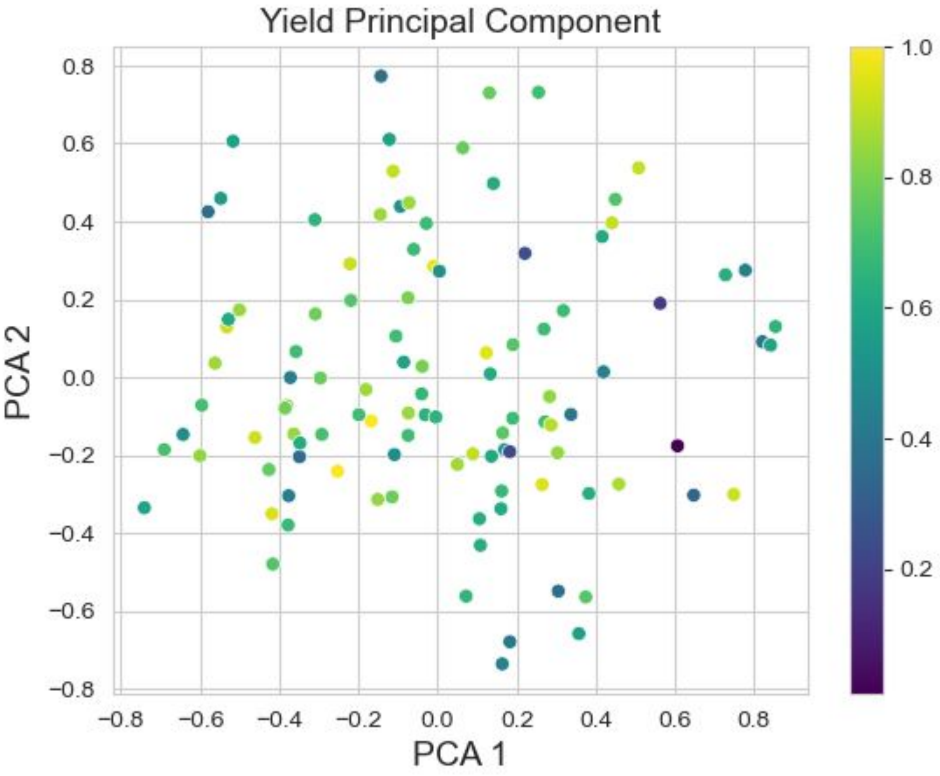
- ❑ CO Flow Rate
- ❑ H2 Flow Rate
- ❑ Pressure
- ❑ Temperature
- ❑ Solvent Fraction
- ❑ Ligand Fraction
- ❑ Olefin Fraction

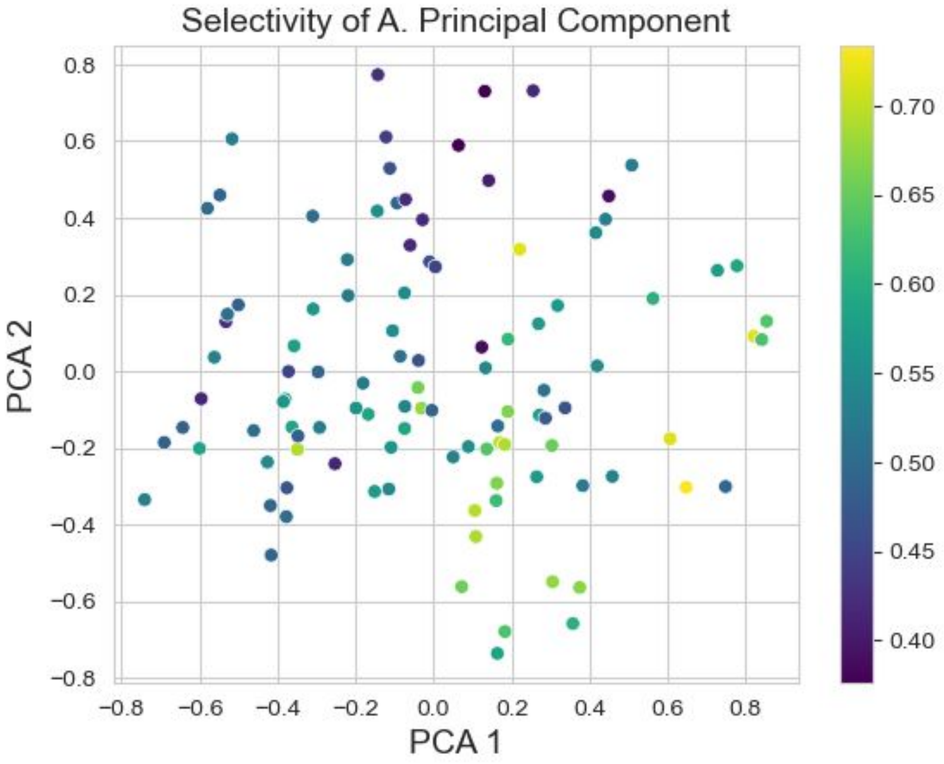
To have better control over:

- ❑ Selectivity (linear vs branched)
- ❑ Yield



Exploratory Data Analysis





Model Development

A total of 6 regression models were developed and evaluated using Bayesian methods. The basis of each model is this formula:

$$Y_i \sim N\left(\alpha_j + \sum_{j=1}^{p} X_{ij}\beta_j + e_i\,, \sigma\right)$$

Model 1: Non quadratic linear regression

Assuming independent normal distributions of Y_1 and Y_2.

Using uninformative priors for σ, α, and β:

$$\sigma \sim \Gamma(0.01, 0.01)\ \alpha \sim N(0, 100)\ \ \beta \sim N(0, 100)$$

Model 2: BLASSO Regression

Assuming bivariate normal distributions of Y_1 and $Y_{2,}$ and variance of τ_e in place of σ.

Using uninformative priors for both σ, β, τ_e, and τ_b :

$$\alpha \sim N(0, 100), \tau_e \sim \Gamma(0.01, 0.01),\ \tau_b \sim \Gamma(0.01, 0.01), and\ \beta \sim DDexp(0, \tau_e \tau_b)$$

Model 3: BLASSO with Random Effects

Same as Model 2 but introducing $\mu_n \sim N(0, 100)$ as the means for α and β.

Model 4: BLASSO using Quadratic Terms

Same as Model 2 introducing quadratic terms for CO, H, Pressure, Temperature, and the Octene-to-Rhodium ratio. Interaction terms were introduced between the gaseous reactants, pressure, and temperature. Basis for interaction is the ideal gas law (PV = nRT) and the fact that pressure and temperature affect the solubility of the gases in the solvent.

Model 5: BLASSO with Yield A/B and full quadratic dataset

Based on Model 4 but instead of yield and selectivity, we use Yield A and Yield B assuming a bivariate normal relationship. Also, instead of select quadratic terms we used second order polynomial terms for all input covariates.

Model 6: BLASSO with precision priors

Using the same transformed inputs and outputs as in Model 5, this model introduced more conditions on prior distributions for the bivariate correlation and removes the assumption that the response variables are independent. The new assumptions are as follows:

$$\tau_{bi} \sim \Gamma(0.25, 0.05)\ where\ i = 1, 2$$

$$\tau_{yi} \sim \Gamma(0.1, 0.1)\ where\ i = 1, 2$$

$$\rho_p \sim \beta(3, 3)$$

$$\rho_{12} = 2\rho_p - 1$$

$$Correlation = \begin{bmatrix} 1 & \rho_{12} \\ \rho_{12} & 1 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} \frac{1}{\sqrt{\tau_{y1}}} & 0 \\ 0 & \frac{1}{\sqrt{\tau_{y2}}} \end{bmatrix}$$

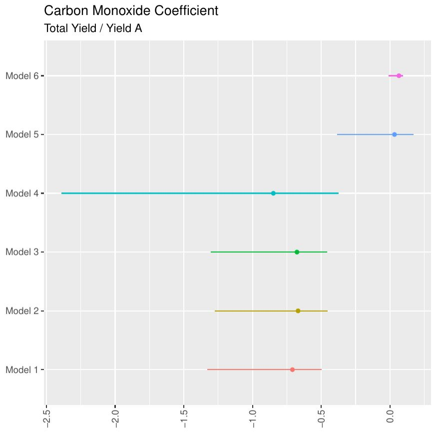
$$\sigma_e = \sigma * Correlation * \sigma$$

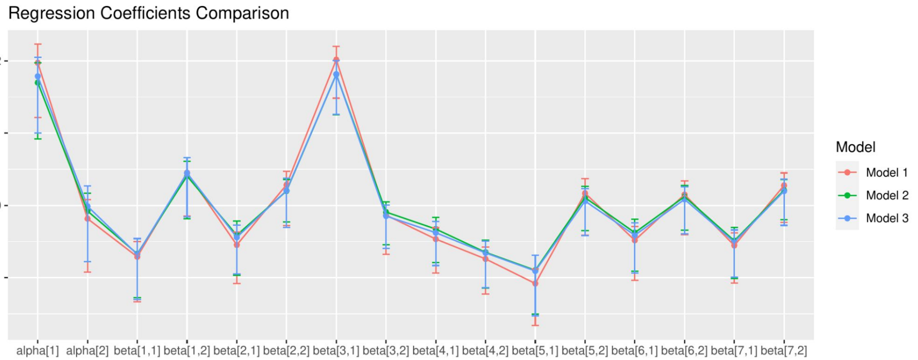
$$\tau_e = \sigma_e^{-1}$$

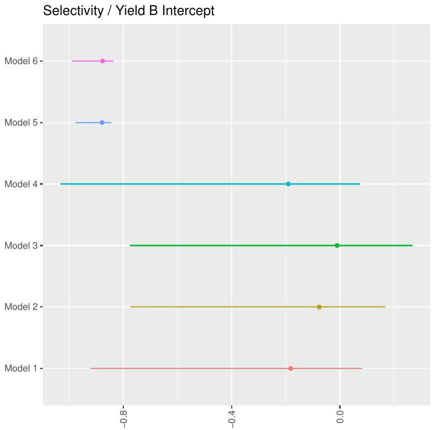
Where, $Y_i \sim N(\mu, \tau_e), \alpha \sim N(0, 10), and\ \beta_{ij} \sim DDexp(0, \tau_i * \tau_{ii})$

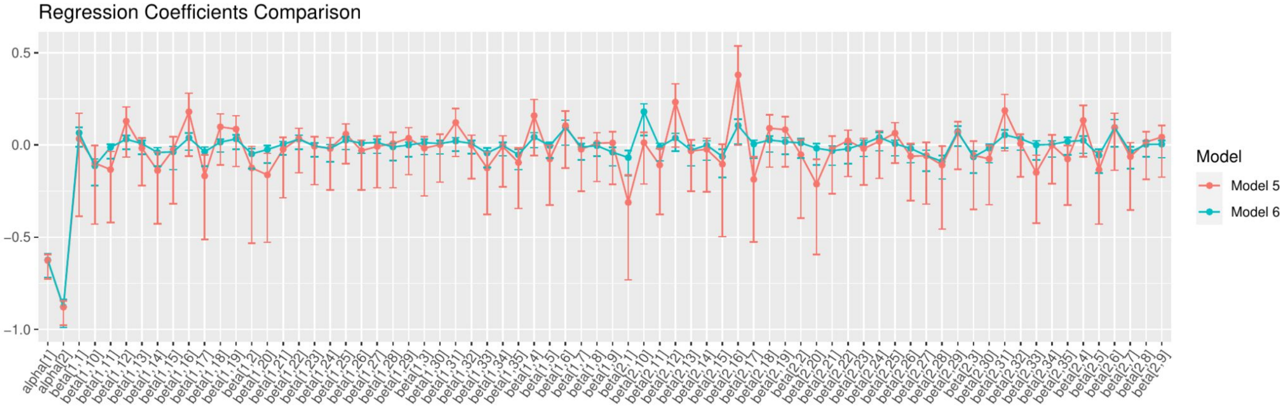
μ in this case it the regression model

Model Results









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Convergence Diagnostics

Summarized Convergence Diagnostics

- The JAGS sampler was setup with a burn-in of 10,000 and 30,000 iterations over 4 chains. Long chains helped establish convergence in lieu of specifying initial values.
- Each model's traceplot is consistent with an algorithm that has converged. Draws oscillate with varying magnitudes around a point and chains overlaps each other. (See Figure 7.1)
- **Autocorrelation:** The Simple model reported having a high correlation ($|\rho| > .50$). The remaining model correlations were below 0.33. The chain length should act to circumvent any concerns regarding high correlation. (See Table 7.1)
- **Geweke:** For the Geweke test, the default measure of the mean of the first 10% vs last 50% were used. Parameters were within tolerable range of $|z| < 2$ for all models with exception to the BLASSO Expanded model where we observed a beta parameter of 13.709 in chain one.
- **Gelman-Rubin**: All models had a Gelman-Rubin statistic that rounded to 1.00 by the nearest hundredth decimal place. The highest Gelman-Rubin statistic was 1.031 found in the BLASSO Expanded model.
- **Effective Sample Size (ESS):** The BLASSO Expanded was the only model that failed the ESS test with a value of 704. (See Table 7.1)

Recap: The Simple model failed at autocorrelation and the BLASSO Expanded failed the Geweke and ESS convergence tests. In consideration of the remaining models that successfully passed convergence testing in visualization and all four diagnostics, the Simple and BLASSO Expanded models are not viable candidates for use.

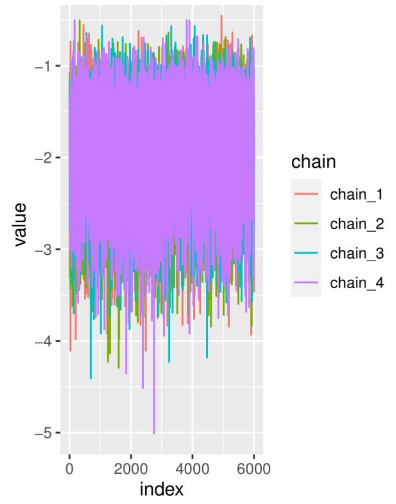


Figure 7.1

Model	Max Corr	Effective Sample Size
Simple	0.61	6539
BLASSO	0.33	6193
BLASSO with RE	0.27	6841
BLASSO Expanded	0.17	704
BLASSO Fully Expanded	0.30	1996
BLASSO Fully Expanded with Correlation	0.19	3256

Table 7.1

Model Selection

The decision is to move forward with the BLASSO Fully Expanded with Correlation model.

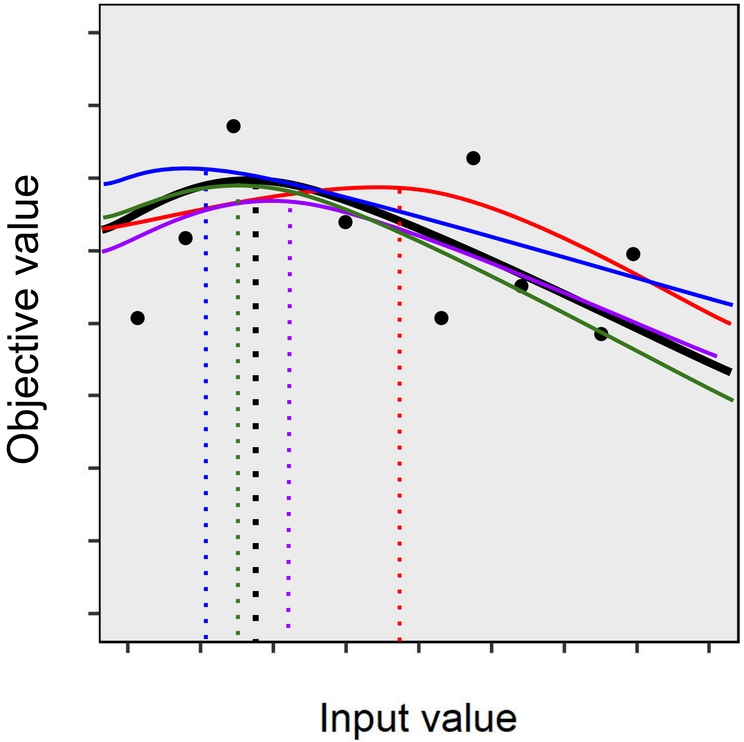
- Our choice in model primarily relied on both information criterion and consideration of the convergence diagnostics results. The Simple and the BLASSO Expanded were our least desirable working models due to their flaws in passing convergence testing.
- The Deviance Information Criterion and Watanabe-Akaike Information Criterion were very close comparatively, as expected. Much of the variability is accounted for in our model of choice and we can see that in the table out below.

Model	WAIC	DIC
Simple	510.0655423	501.6714762
BLASSO	509.1106364	500.4578875
BLASSO with RE	508.6528722	506.1302029
BLASSO Expanded	503.5721781	491.3630566
BLASSO Fully Expanded	387.2356009	357.5594918
BLASSO Fully Expanded with Correlation	253.9805922	237.5108891

Table 8.1

Optimization Process

Simplified example



General process

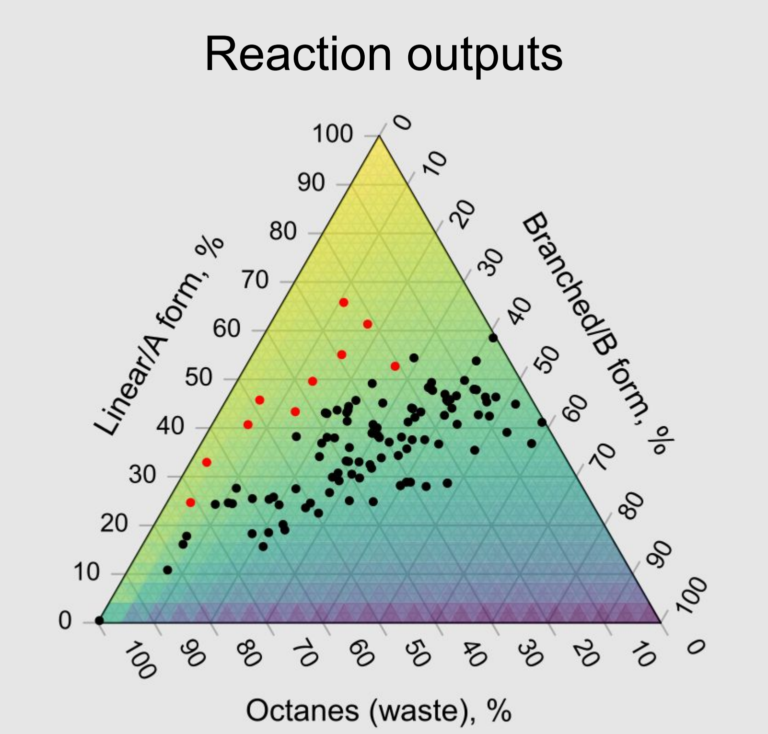
To optimize the reaction, loop until reaction outputs are favorable:

1. Prepare the model with all available data
2. Sample from the posterior distribution of model parameters*
3. For each sampled parameter vector, identify which input values maximize the objective
   - Transforms each sampled parameter vector into a sample from the posterior distribution of inputs that are optimal under the model
4. Collect more data by re-running the reaction with inputs set to sampled values

*Especially extensive thinning can be needed because one wants the thinned MCMC samples to be fully i.i.d., in order to propose reaction inputs that are i.i.d. samples from the posterior distribution of optimal inputs.

Results

With multiple inputs and outputs, this reaction is more complicated than the simplified example, but the same process applies!



- **Objective:** Get outputs into yellow region
- **Black points:** Data collected by experimenters before our analysis
  - Bordering, but not in yellow region
- **Red points:** Proposals from our model
  - Ten points show expected reaction outputs* for 10 candidate sets of reaction inputs/settings
  - From border into yellow region

*IE, posterior expected values of the outputs at the candidate input values.

Discussion

Statistical Precedents

Response surface methodology

- Finding good settings for a process
  - Common problem, long history
  - NC State's Gertrude M. Cox[1], others
- Standard frequentist approach
  - Small *n* scenario, yet asymptotic theory
  - Does not fully capture uncertainty
  - To avoid $p \gtrsim n$, starts with linear model, then awkwardly switches to quadratic*
- Bayesian advantages
  - Fully captures uncertainty, with small *n*
  - Shrinkage priors handle $p \gtrsim n$

Bayesian bandit

- Resembles Thompson sampling[2]
- Experiment settings selected in proportion to the model's posterior probability that they are optimal
- Balances exploration and exploitation

[1]Cochran WG, Cox MG. Experimental designs. Chapter 8A. John Wiley & Sons; 1957.

[2]Thompson WR. Biometrika. 1933;25:285-94.

*An at-least quadratic model is needed because, under a linear model, input values that maximize the response can only be boundary values.[1]

~~Limitations~~ …Opportunities

- Experiment still in-progress as of this presentation
- Link with prior literature
  - How does our proposed approach perform in comparison with previous Bayesian attempts at reaction optimization?[1,2]
- Develop the theory
  - We lack optimality proofs or bounds for our approach
  - Could they be obtained by extending proofs on Thompson sampling?
- Improve model flexibility
  - We used quadratic polynomial regression (including all 2-variable interactions)
  - Splines or gaussian processes would increase flexibility
  - They also tend to extrapolate better outside observed data
  - However, they add complexity, especially for interaction terms
  - Additionally, their greater flexibility may increase sensitivity to priors

[1]Clayton AD et al. Algorithms for the self-optimisation of chemical reactions. Reaction Chemistry & Engineering. 2019;4:1545-54.
[2]Shields BJ et al. Bayesian reaction optimization as a tool for chemical synthesis. Nature. 2021;590:89-96.